

JEE MAIN REVISION MASTER: ATOMS AND NUCLEI SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 284 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Atoms And Nuclei - Core Application	284 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Atoms And Nuclei - Core Application (Very High Yield — 284 PYQs)

Core Principle: Key problem pattern in Atoms And Nuclei.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "The ratio of momentum of the photons of the 1st and 2nd line of Balmer series of Hydrogen atoms is α/β " [Ans: D]
- [2026 Evening Shift] Q#1 (MCQ): "Which of the following pair of nuclei are isobars of the element?... " [Ans: A]
- [2026 Evening Shift] Q#1 (Numerical): "The average energy released per fission for the nucleus of $_{92}^{235}\text{U}$ is 190 MeV . When all the atoms of 47 g pure $_{92}^{235}\text{U}$ underg... " [Ans: 228]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

The ratio of momentum of the photons of the 1st and 2nd line of Balmer series of Hydrogen atoms is α/β . The possible values of α and β are:-

- (A) 27 and 20
- (B) 3 and 16
- (C) 5 and 36
- (D) 20 and 27 **[Correct]**

Correct Answer: D

Solution Analysis: According to the de Broglie relation, the momentum (p) of a photon is inversely proportional to its wavelength (λ): $p = (h / \lambda)$. Therefore, the ratio of momenta is the inverse ratio of their wavelengths:

$\frac{p_1}{p_2} = \frac{\lambda_2}{\lambda_1}$ The Balmer series occurs when an electron transitions from a higher energy level (n_2) to the second energy level ($n_1=2$). The wavelength is given by Rydberg formula: $(1 / \lambda) = R(\frac{1}{2^2} - \frac{1}{n^2})$ For 1st line of Balmer transition from $n=3$ to $n=2$. $\frac{1}{\lambda_1} = R(\frac{1}{2^2} - \frac{1}{3^2})$